

Cross-Lingual Journalism Intelligence Pipeline

Full technical specification of the multi-source, multi-lingual summarization system — ingestion, translation, clustering, synthesis, section taxonomy, source attribution, hover interface, and source metadata layer.

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1. System Overview & Design Principles

This system ingests news articles and short-form social content from multiple sources in multiple languages, processes them through a structured NLP pipeline, and produces a single coherent journalistic summary divided into epistemic sections. Every sentence in the output is traceable to the exact original text of its source via a hover interface.

Core Design Principles

- Translation happens at sentence level with explicit original↔translation mapping — originals are never discarded.
- The LLM only synthesises — it never generates facts. All factual content is extracted from sources upstream.
- Volume is managed by extractive compression and cross-source deduplication before any LLM call.
- Short-form content (Telegram, X) enters the same pipeline but is never compressed and can support but not solely establish a verified fact.
- Clusters are built around event identity, not publication date.
- Re-synthesis is an incremental integration operation, not a full regeneration.
- Source credibility and outlet metadata are surfaced to the user as context, not hidden.

2. Content Ingestion

Four content types are ingested. Each receives a type tag, a credibility level, and a language code at the moment of arrival. These tags travel through the entire pipeline.

Type	Examples	Compression	Credibility default
Article	Al Jazeera, Reuters, Le Monde	35–40% extractive	outlet-level
Short post	Telegram channels, X posts	None — kept whole	channel-level
Statement	Translated minister / president quote	None — kept whole	official
OSINT	Anonymous channels, field reporters	None — kept whole	unverified

Short posts and statements skip sentence splitting entirely and enter the pipeline as single units. A short post from an official verified channel (e.g. Al Jazeera Telegram, a presidential account) carries higher credibility than an anonymous OSINT channel. This credibility weight influences section classification downstream — an exclusive claim from an official channel can be elevated to the Official Statements section even if unconfirmed by other outlets.

3. Language Handling

The pipeline strictly separates operations that must run on native-language text from those that run after translation. This boundary is the architectural foundation of source attribution integrity.

3.1 Before Translation — Native Language Operations

- Language detection: langdetect + lingua-py. Detects language code per article (arb_Arab, fra_Latn, eng_Latn, spa_Latn, etc.).
- Sentence splitting: spaCy xx_sent_ud_sm (multilingual) for articles. Short posts skip this step.
- NER on original text: Davlan/xlm-roberta-large-ner-hrl. Runs on native text — entity names extracted here are unaffected by transliteration errors that translation can introduce. Covers Arabic, French, English, Spanish natively.

3.2 The Translation Boundary

Translation runs sentence by sentence via facebook/nllb-200-distilled-1.3B (200 languages). English articles skip the translation call entirely — language detection at step 1 routes them directly. Every translated sentence is immediately paired with its original before any further processing:

Sentence Mapping Object

- original_text — verbatim from source in native language
- original_lang — NLLB language code (arb_Arab, fra_Latn...)
- translated_text — English output from NLLB
- article_id — source article identifier
- outlet — publication name
- url — canonical source URL
- char_start / char_end — byte offsets in original article for hover retrieval
- content_type — article | short_post | statement | osint
- credibility — official | outlet-level | unverified

From this point forward all models receive English text. The original is preserved in the mapping object and is never touched again by any model. It surfaces only at the hover layer when the journalist requests the source.

4. Event Fingerprint Matching & Cluster Management

Clustering is built around event identity, not publication date. This solves the late-publication problem: an article published today about an event from three days ago correctly joins the three-day-old cluster because the entities and event match, not because of timestamp proximity.

4.1 Event Fingerprint

Each cluster has a fingerprint generated from its seed article and updated as new content arrives. The fingerprint contains: primary named entities (persons, organisations, locations from NER), event action verb (strike, announcement, arrest...), and event date (extracted from article metadata or NER time entities).

4.2 Cluster Routing Logic

New event	No matching cluster above similarity threshold. New cluster created. Synthesis waits for minimum 2 independent sources before running.
Active cluster	Matching cluster exists and is under 72 hours old. Content joins cluster. Re-synthesis flagged.
Closed cluster	Matching cluster exists but is over 72 hours old. Content enters follow-up queue. A follow-up section is appended — main synthesis is not modified.
Ongoing story	Parent thread (e.g. Russia-Ukraine war) with child clusters per event. New events create child clusters that inherit parent context without polluting the parent synthesis.

Time functions as a decay factor, not a hard boundary. The 72-hour threshold is configurable per topic category — breaking news clusters may close in 24 hours while investigative threads may stay open for weeks.

5. Compression, Deduplication & Confirmation Counting

This stage runs entirely without an LLM. Its purpose is to reduce the input to the synthesis model from hundreds of sentences to a small structured set of unique fact objects with confirmation metadata. The volume reduction comes primarily from cross-source deduplication, not from cutting within individual articles.

5.1 Intra-Article Compression (Articles Only)

- Sentences scored by: presence of named entities, numerical values, direct quotes, attribution language (said/confirmed/denied/stated).
- Top 35–40% of sentences kept. Investigative or analysis articles use a higher ceiling of 60–70%.
- Short posts, statements, and OSINT posts are never compressed — they are already minimal.
- Compression selects original sentences — no rewriting. Attribution chain is preserved.

5.2 Cross-Source Deduplication & Fact Objects

After compression, all sentences from all sources in a cluster are compared against each other. Semantically equivalent sentences are merged into a single fact object. The confirmation count and source list travel into the LLM prompt.

Fact Object Structure

- fact — representative sentence in English
- first_seen — outlet and original sentence that established the fact
- confirmed_by — list of outlets whose compressed sentences matched this fact
- confirmation_count — integer count of confirming outlets
- short_post_confirmations — Telegram/X posts that corroborate
- contradicted_by — outlets whose sentences conflict with this fact
- unique_to — outlet name if confirmation_count is 1 (exclusive flag)
- section_flag — pre-classification from mDeBERTa (verified_fact, exclusive, etc.)

A fact confirmed by 7 outlets is represented by one sentence in the LLM input. The LLM sees: 'Russian missiles struck residential areas — confirmed by 7 outlets: Reuters, AFP, Al Jazeera, Le Monde, BBC, AP, Mayadeen.' This becomes both the synthesis input and the data behind the hover confirmation badge.

6. Section Taxonomy & Classification Pipeline

Every fact object is classified into one or more sections before reaching the LLM. Classification uses mDeBERTa-v3-base-mnli-xnli via zero-shot NLI on the translated English text. Section labels are defined in plain English — no fine-tuning required.

Section	Definition	Epistemic Status
Verified Facts	Documented, undeniable events confirmed by multiple sources (official posts, confirmed strikes, published decrees)	<i>Certain</i>
Official Statements	Direct quotes from governments, institutions, spokespeople, ministers	<i>Certain as stated</i>
Historical Context — Agreed	Background accepted across sources without dispute	<i>Established</i>
Historical Context — Disputed	Competing narratives on causes, responsibility, or framing	<i>Contested</i>
Exclusive / Single-Source	Claims appearing in only one outlet, not corroborated by any other source	<i>Unverified</i>
Analysis & Interpretation	Journalist or expert reading of events, with detected bias frequency	<i>Subjective</i>
Numbers & Data	Statistics, polls, financial figures, casualties — all numerical claims	<i>Verifiable</i>
Reactions & Positions	How different actors (states, parties, organisations) responded to the event	<i>Attributed</i>
What Remains Unknown	Gaps explicitly acknowledged across sources — what no outlet addresses	<i>Epistemic gap</i>
Timeline	Chronological sequence of events with timestamps	<i>Structural</i>
Legal & Accountability	Charges, court decisions, sanctions — factual but legally distinct	<i>Certain</i>
Officials Said vs Data	Diff section: when official statements contradict measurable figures	<i>Contested</i>
Follow-Up	Articles arriving after 72h window on a closed cluster	<i>Historical update</i>

6.1 Classification Rules

- Short posts from official verified channels are always classified as Official Statements or Verified Facts — credibility tag overrides model confidence.
- OSINT posts are always classified as Exclusive/Single-Source regardless of content.
- Any fact with `confirmation_count = 1` is flagged exclusive regardless of classifier output.
- Sentences containing quotation marks or attribution verbs (said, stated, confirmed, denied) always land in Official Statements.
- Contradicted facts receive a dual classification: their original section plus a contradiction flag.

7. Bias Detection Layer

Bias detection runs on each translated sentence using `valurank/distilroberta-bias`. Results are aggregated per outlet and per cluster to produce a bias frequency map that surfaces in the Analysis section of the synthesis output.

Bias Outputs

- `bias_score` per sentence (0.0 – 1.0)
- `bias_direction` per outlet: left | right | neutral | institutional | nationalistic
- `loaded_terms`: specific words or phrases flagged as carrying implicit framing

- framing_technique: word_choice | omission | emphasis | false_balance | none
- cluster-level bias_frequency_map: percentage of sentences per direction across all outlets

The bias frequency map is not a judgment. It is a frequency indicator the journalist interprets. If 60% of sentences across all sources carry institutional framing, that is surfaced as a signal. The Analysis section of the synthesis output contains this map as a visual distribution alongside the synthesised analytical content.

8. LLM Synthesis

The LLM receives a structured digest of fact objects, not raw text. Its role is curation and connection — choosing which facts to surface in which section, writing neutral journalistic connectors, and flagging contradictions. It never generates factual content independently.

8.1 LLM Input Structure

Structured Prompt Components

- CONFIRMED FACTS: fact objects with confirmation counts and outlet lists
- SHORT POST SIGNALS: Telegram/X corroborations with credibility tags
- EXCLUSIVE / SINGLE SOURCE: facts flagged unique_to a single outlet
- OFFICIAL STATEMENTS: attributed quotes from officials and institutions
- CONTRADICTIONS DETECTED: pairs of conflicting facts from different outlets
- EXISTING SYNTHESIS (re-synthesis only): previous output as base to integrate into
- BIAS SUMMARY: cluster-level bias frequency map from distilroberta
- SECTION INSTRUCTIONS: explicit list of sections to produce

8.2 First Synthesis

Runs when a cluster reaches its minimum source threshold (2 independent sources). The LLM produces the full section-divided output. Each sentence references the fact object indices it was derived from. Confirmation counts appear inline: 'confirmed by 5 outlets' rather than repeating source names in the body.

8.3 Incremental Re-synthesis

Runs when new content joins an active cluster. The previous synthesis is included in the prompt. The LLM integrates new facts into existing sections, updates confirmation counts where facts are now corroborated by additional sources, adds new exclusive facts to the exclusive section, and surfaces any new contradictions. The structure is preserved. This is an integration operation, not a regeneration — output grows incrementally.

8.4 Follow-Up Synthesis

Runs when content arrives for a closed cluster (over 72 hours old). The LLM receives only the existing synthesis and the follow-up fact objects. It appends a follow-up section without modifying the original synthesis, preserving the historical record of initial coverage.

8.5 Models

- Primary: Qwen/Qwen2.5-7B-Instruct — best structured JSON output at 7B, strong instruction following, 32k context window.
- Alternative for Arabic-heavy pipelines: CohereForAI/aya-expense-8b — 23 languages including Arabic and French, handles code-switching.
- Hardware note: both models require quantization (Q4_K_M GGUF via Ollama) for local inference on 4GB VRAM. CPU inference via 32GB RAM at ~2–4 tokens/second.

9. Source Attribution & Hover Interface

After synthesis, each output sentence is matched against the fact objects via vector search. Since each fact object carries its source list and original text pointers, attribution is pulled from the fact object rather than re-searched from scratch. This is faster and more precise.

9.1 Attribution Locking

- Each synthesis sentence is embedded using BAAI/bge-m3.
- Matched against the cluster's translated sentence pool in Qdrant at similarity threshold 0.82.
- Each match returns a sentence mapping object containing the original text, language, outlet, URL, and char offsets.
- Multiple matches per sentence are kept — one synthesis sentence can point to quotes from 3 different outlets.

9.2 Hover Popover Data

Per Sentence Hover Payload

- `summary_sentence_id` — reference to the output sentence
- `sources[]` — list of source objects, one per matched outlet:
 - outlet name, article title, publication URL, `published_at` timestamp
 - `original_text` — verbatim in native language (Arabic, French, etc.)
 - `translated_text` — English translation for user reference
 - `char_start` / `char_end` — position in original article
 - `verified`: true/false — was the original text found verbatim at these offsets
 - `credibility` — official | outlet-level | unverified
 - `confirmation_count` — total number of outlets confirming this fact
 - `contradiction_note` — description if sources frame the fact differently

9.3 Frontend Display

- Summary sentences rendered as hoverable spans with a dashed underline indicating attribution exists.
- Hover triggers a popover showing all source quotes stacked, each with outlet name, timestamp, original language text, and English translation.
- Contradictions between sources shown with a warning flag and a side-by-side diff of the conflicting phrasings.
- Confirmation count shown as a badge: 'confirmed by 5 outlets' — clicking expands the full outlet list.
- Exclusive facts shown with a distinct visual treatment and a single-source warning.
- Official statements shown with the official badge and direct link to source.

10. Source Metadata System — Admin & User Layer

Every outlet in the system has an admin-managed metadata profile. This profile is surfaced to the journalist as contextual background when they hover on a source name, giving editorial context without editorialising in the synthesis itself.

10.1 Outlet Profile Fields (Admin-Managed)

Outlet Metadata Object

- name — canonical outlet name
- country — country of registration
- ownership — parent company, owner name, or state entity
- funding_source — private, state-funded, publicly funded, NGO
- political_leaning — left | centre-left | centre | centre-right | right | state | religious | nationalist
- known_biases[] — specific documented editorial positions or conflicts of interest
- credibility_tier — tier 1 (major verified international) | tier 2 (regional verified) | tier 3 (unverified/OSINT)
- languages[] — languages the outlet publishes in
- fact_check_record — known corrections or retracted stories (manually curated)
- notes — free text for editorial context (e.g. 'owned by Qatari state', 'Russian state media')
- last_updated — date of last admin review

10.2 Admin Interface

The admin panel allows the editorial team to create, update, and review outlet profiles. Profiles are not generated automatically — they are manually curated to ensure accuracy. An outlet with no profile defaults to credibility tier 3. Profile completeness is shown to the journalist so they know when context is partial or absent.

10.3 User-Facing Display

When a journalist hovers on a source name in the hover popover, a secondary panel shows the outlet profile. The display is informative and non-prescriptive — it states facts about the outlet (owned by X, funded by Y, known positions on Z) without telling the journalist how to interpret the information. Political leaning and ownership are shown as factual metadata, not as editorial warnings.

11. Full Model Stack & Hardware Notes

Task	Model (HuggingFace)	Language	Note
Language detection	langdetect + lingua-py	Native	Runs on CPU, negligible cost
Sentence splitting	spaCy xx_sent_ud_sm	Native	Multilingual, CPU
NER on originals	Davlan/xlm-roberta-large-ner-hrl	Native	AR, FR, EN, ES — GPU recommended
Translation	facebook/nllb-200-distilled-1.3B	Any→EN	Load float16 for 4GB VRAM
Embeddings	BAAI/bge-m3	EN (post-translate)	GPU, 570MB VRAM
Section classification	MoritzLaurer/mDeBERTa-v3-base-mnli-xnli	EN	Zero-shot NLI, GPU
Bias detection	valurank/distilroberta-bias	EN	CPU viable, 80MB
Synthesis + re-synthesis	Qwen/Qwen2.5-7B-Instruct	EN	Q4_K_M via Ollama, CPU 32GB RAM
Synthesis alt (Arabic-heavy)	CohereForAI/aya-expanse-8b	EN	Swap for Qwen if Arabic dominates
Vector storage	Qdrant (self-hosted)	EN embeddings	Local, no cloud dependency

11.1 Hardware Allocation (HP ZBook Studio G7 — 32GB RAM, Quadro T1000 4GB)

GPU (4GB VRAM)	BAAI/bge-m3, mDeBERTa, distilroberta-bias, xlm-roberta NER, NLLB-1.3B in float16. All small models run here.
CPU + 32GB RAM	Qwen2.5-7B Q4_K_M via Ollama. Runs at ~2–4 tokens/second. Viable for background batch processing.
Storage	~10GB for all models. Current free space (~29GB) is sufficient but tight. Free at least 50GB before deployment to accommodate Qdrant growth.

If synthesis speed becomes a bottleneck, the Qwen call can be offloaded to the HuggingFace Inference API free tier. All small classification and embedding models remain local. This hybrid approach keeps costs at zero while removing the CPU inference bottleneck for the single heaviest step.

12. Complete Pipeline Flow Reference

The following table traces every step from raw ingestion to final output. Language column shows what language the data is in at that step.

INGESTION	Scrape articles, Telegram posts, X posts, statements. Assign type tag and credibility level.
LANG DETECTION	Detect language code per piece. langdetect + lingua-py. Native language. Short posts detected as single unit.
SENTENCE SPLIT	Articles: split into sentences via spaCy xx_sent_ud_sm. Short posts/statements: skip, treated as single sentence. Native language.
NER (native)	Named entity recognition on original text. Davlan/xlm-roberta-large-ner-hrl. Outputs: persons, orgs, locations, times. Native language.
TRANSLATION	Sentence-level translation via NLLB-200-1.3B. English articles skip this call. Each translated sentence immediately paired with original in mapping object.
FINGERPRINT MATCH	Match incoming content against existing cluster fingerprints using NER entities + event markers. Route to: new cluster / active cluster / follow-up queue.
COMPRESSION	Articles only: score sentences, keep top 35–40%. Short posts/statements untouched. Runs on translated English text. No LLM.
DEDUPLICATION	Cross-source comparison of compressed sentences within cluster. Merge semantically equivalent sentences into fact objects. Build confirmation counts. English.
CLASSIFICATION	mDeBERTa-v3-base-mnli-xnli zero-shot NLI on translated sentences. Assigns section label per fact object. English.
BIAS DETECTION	valurank/distilroberta-bias per sentence. Aggregate per outlet and per cluster. Build bias_frequency_map. English.
EMBEDDING	BAAI/bge-m3 embeds all translated sentences. Stored in Qdrant with mapping object metadata. English embeddings.
LLM INPUT BUILD	Construct structured prompt: confirmed facts with counts, short post signals, exclusives, statements, contradictions, previous synthesis (if re-synthesis).
LLM SYNTHESIS	Qwen2.5-7B-Instruct (or aya-expanse-8b). Produces section-divided output. First synthesis, incremental re-synthesis, or follow-up append. English output.
ATTRIBUTION LOCK	Each output sentence embedded and matched against cluster sentence pool in Qdrant. Attribution payload assembled per sentence from fact objects and mapping objects.
FOLLOW-UP APPEND	If follow-up queue non-empty: light re-synthesis appends follow-up section. Original synthesis preserved as historical record.
API OUTPUT	FastAPI returns structured JSON: section-divided summary, per-sentence hover payloads (original text, translation, outlet metadata), bias map, cluster metadata.
FRONTEND	React. Section blocks. Hoverable sentence spans. Popover: original language quote + translation + outlet profile. Bias distribution. Confirmation badges.

Output Structure per Cluster

- cluster_id, topic, event_date, synthesis_generated_at, source_count

- sections{}: verified_facts[], official_statements[], historical_agreed[], historical_disputed[], exclusive[], analysis[], data[], reactions[], unknown[], timeline[], legal[], officials_vs_data[], follow_up[]
- per sentence: text, section, sources[] {outlet, original_text, translated_text, url, verified, credibility}, confirmation_count, contradiction_note, bias_flag
- bias_distribution{}: direction → frequency map across all outlets
- coverage_gaps[]: facts acknowledged as unknown across sources
- outlet_profiles{}: admin-managed metadata per outlet shown on hover